NSW Public Transport: Problem Diagnosis and a Portfolio of GTFS-R Augmented Reality Project Ideas

TL;DR

- NSW public transport in 2024 to 2026 is defined by a reliability slump (Sydney Trains suburban on-time running fell to 83.5% in FY2025, well below the 92% target, and just over 70% of intercity services ran on schedule, a ten-point drop in a single year), a persistent bus driver shortage that forced timetable cuts, and a deep spatial inequity where all eight Western Sydney LGAs sit in the bottom third of Sydney for public transport access.
- The Transport for NSW Open Data Hub publishes rich, well-documented GTFS and GTFS-Realtime feeds (Trip Updates, Vehicle Positions, Service Alerts) across buses, ferries, light rail, Sydney Trains, NSW Trains, Metro and regional buses, updated every 15 seconds, plus complementary datasets (Opal patronage, Car Park API, Live Traffic Hazards, Trip Planner APIs, facilities and accessibility data, cycleways) that can be joined by shared stop IDs.
- This report proposes twelve concrete projects, each using live GTFS-R as a core input and one of three physical AR setups (a fixed 5km x 4km Sydney CBD scale model, a Kinect AR sandbox, or a LEGO participatory grid), spanning reliability diagnostics, equity mapping, disruption response, and community consultation.

Key Findings

Reliability has deteriorated across nearly every mode. In FY2025 only 83.5% of Sydney Trains suburban services ran on time (within five minutes), a drop of more than five percentage points from FY2024 and far below the 92% target last met in FY2022. Just over 70 percent of intercity services ran on schedule, a ten-point drop in one year, against a 92% within-six-minutes target that was also last exceeded in FY2022. The T1 North Shore & Western Line and the T9 Northern Line, the longest lines, both recorded about 79% punctuality, with the T9 nearly eight percentage points below its prior-year result. The 20 May 2025 overhead wiring failure at Homebush caused two days of network-wide chaos and triggered an Independent Rail Review led by Dr Kerry Schott AO with Carolyn Walsh and Trevor Armstrong (final report 29 August 2025). The review found the overhead wire that failed had been identified as a risk in 2020 but was never entered into Sydney Trains' defect management system and was not repaired. The government accepted all 12 recommendations and committed $458.4 million over four years, including a shift from time-based to risk-based maintenance and "critical maintenance zones" starting with the Homebush to Strathfield corridor. (NSW Government)

The bus driver shortage is the single biggest driver of bus service failures. The NSW Bus Industry Taskforce found most cancellations were a direct result of driver shortages. The Taskforce Summary of Actions records the vacancy rate halving to 249 by February 2024 compared with 502 in February 2023, and TfNSW reported 154 vacancies across metropolitan Sydney by 3 March 2025. Operators including Busways continued to cut timetables into late 2025 and early 2026 (North Shore and West regions, a shortage of about 50 drivers). The 2025

NSW Audit Office report on bus contracts found operators met the customer complaints KPI only 41.6% of the time between January 2023 and May 2024, down from 85.2% in the earlier period. Nsw

Spatial inequity in Western Sydney is structural. All eight Western Sydney LGAs are in the bottom third of Sydney for public transport access, while the best-served third are all in the East or North. A NSW Legislative Council inquiry (Portfolio Committee No. 6), reporting 29 April 2024, made 20 recommendations and urged the government to build public transport before approving more housing on the outer fringes. A separate November 2024 inquiry into transport supporting Western Sydney International Airport warned the region is car-dependent and that insufficient connections could worsen congestion, recommending rapid buses from Penrith, Liverpool and Campbelltown in time for the airport's late-2026 opening. Jo Haylen MP nsw

Major growth is arriving ahead of transport. Sydney Metro City opened 19 August 2024 and in its first year recorded more than 66.8 million journeys, with an average 99.4% on-time running and roughly 210,000 trips a weekday (about 120,000 of them under Sydney Harbour). The Sydenham to Bankstown conversion is due 2026. Sydney Metro West (Westmead to Hunter Street, ten stations) is due 2032. Sydney Metro Western Sydney Airport (St Marys to the Aerotropolis, six stations, 23km) is timed for the airport opening. The Aerotropolis is overwhelmingly zoned for employment, raising concerns about key-worker housing and feeder transport. The Fifth Estate

Overcrowding concentrates on specific corridors. The overcrowding benchmark is a load of 135% (all seats filled plus standing). Sydney Trains data reported by the Sydney Morning Herald shows the T1 Western Line as the busiest morning line with an average load of 145% between 8am and 9am, followed by the T1 Northern via Strathfield (137%) and T4 Illawarra (132%). A double-deck eight-car train has a practical crush capacity of around 1,750 to 2,100.

Bus priority works where deployed. TfNSW's own case study reports a 15% reduction in overall bus travel times on the T80 Liverpool to Parramatta Transitway in FY2022, using SCATS, the Signal Priority Engine and the PTIPS vehicle tracking system to grant priority green signals; the rapid route covers 31km through 68 intersections and carries over two million passengers a year.

Zero emission bus transition is underway but slow. NSW aims for a fully zero-emission fleet by 2035 in Greater Sydney, 2040 in outer metro, and 2047 regionally. TfNSW reports 132 ZEBs already in service against a fleet of over 8,000; Brookvale depot, converted with gantry-mounted pantograph fast charging and a 250kW rooftop solar system, will accommodate 229 battery electric buses by the end of 2028, making it Australia's largest electric bus depot. Transport for NSW

Data science on TfNSW GTFS-R is already established. The University of Sydney TransportLab SCALUT project built a GTFS data pipeline used to validate TfNSW's transit signal priority (PTIPS), and published 25 months of actual bus arrival and departure times for Greater Sydney in Nature Scientific Data. An iMOVE-supported PhD project uses archived GTFS to link road design to bus performance across Greater Sydney.

Details: The TfNSW Open Data Landscape

GTFS and GTFS-Realtime. TfNSW publishes static GTFS (the Timetables Complete GTFS bundle, plus a "For Realtime" subset) and three GTFS-R entity types via `https://api.transport.nsw.gov.au/v1/gtfs/realtime/`: Trip Updates, Vehicle Positions and Service Alerts. Feeds cover buses, ferries, light rail, Sydney Trains, NSW Trains, Metro and regional buses. Real-time data updates every 15 seconds; static data daily. Access uses an API key (`Authorization: apikey`). A TfNSW proto file (extension 1007) is required to decode carriage and vehicle descriptor extensions.

Known quirks. The ferry Vehicle Positions feed reports current and future positions, creating a "trail" of ghost vehicles unless filtered by current trip. Bus trip IDs change when a route is temporarily altered. Light rail GTFS is updated ad hoc (a manual process). Some regional feeds (for example Nowra Coaches / Sydney Surrounds) have had unresolved trip ID and timestamp defects. The buses feed is a single combined feed for all metro and outer-metro operators, requiring client-side filtering by operator. A v2 feed suite adds predictive trip and carriage-level occupancy for Sydney Trains (`departure_occupancy_status` and per-carriage predictive occupancy), and Waratah trains supply actual occupancy in the vehicle positions feed.

Complementary datasets that join to GTFS-R (via shared TSN / parent_station stop IDs): the Car Park API (real-time Park&Ride occupancy including Metro stations Tallawong, Bella Vista, Hills Showground, Cherrybrook and Kellyville, updated in real time); Live Traffic Hazards API (incidents, roadworks, floods, major events as GeoJSON at `https://api.transport.nsw.gov.au/v1/live/hazards/`); Trip Planner APIs (Stop Finder, Trip Planner, Departure Board, Service Alerts, Coordinate Request; default 60,000 calls a day); Opal Patronage (daily by mode since January 2020, updated monthly); Opal Tap On and Tap Off (Release 3, 2020, includes origin-destination by travel zone); Public Transport Location Facilities and Operators (daily, station facilities, joins to GTFS by parent station ID); GTFS Pathways (step-free navigation inside stations, bundled into Timetables Complete since June 2023); DSAPT accessibility compliance; Infrastructure Cycleway Data (monthly); and Cycling Count.

The best 5km x 4km model area is the Sydney CBD spine (Central to Circular Quay, taking in Town Hall, Wynyard, Barangaroo and Martin Place). It is the only zone where all five modes converge: heavy rail (T1, T2, T3, T4, T8, T9 through the City Circle), Metro M1 (Central, Gadigal, Martin Place, Barangaroo), light rail (L2 and L3 on George Street), ferries (Circular Quay and Barangaroo) and bus interchanges (Wynyard and Circular Quay). It contains the three busiest stations in NSW: Central (around 270,000 customers a day), Town Hall (about 133,000 a day) and Wynyard (about 92,000 a day). Parramatta ranks fourth but is essentially a heavy-rail and bus interchange with separate light rail and river ferry, so it does not concentrate all modes in one tight cluster. That density makes the CBD the richest subject for a fixed model.

The Project Portfolio

Setup A: The fixed 5km x 4km Sydney CBD scale model (projection mapping)

1. Pulse CBD: Live Multimodal Network Heartbeat A projection-mapped scale model of the Sydney CBD showing every train, metro, light rail vehicle, ferry and bus moving in real time as points of light along their corridors, with colour encoding on-time status (green on time, amber late, red cancelled or held). Tackles the reliability slump and the public's difficulty seeing how disruption cascades between modes. Target users: museum and visitor-centre audiences, TfNSW communications teams, school groups. The medium: vehicle positions are projected as moving dots onto the physical streets and rail corridors of the model; a wall panel or tablet shows aggregate on-time percentages. Interaction loop: visitors touch a station on the model (capacitive markers) to spotlight services through it. Data: GTFS-R Vehicle Positions and Trip Updates for all modes, static GTFS for geometry, Service Alerts for incident overlays. Analysis: map-matching vehicle positions to route shapes, computing schedule deviation per vehicle from Trip Updates. Feasibility: highly achievable; the main effort is accurate projection calibration to the model and a robust feed ingester. A good flagship demonstrator.

2. Cascade: Rail Incident Ripple Simulator When a real incident is detected (or replayed from an archive such as the Homebush 20 May 2025 event), the model animates how delay propagates outward from the failure point across shared track sections and interchanges. Tackles rail incident cascades and the Independent Rail Review finding that recovery is too slow and communication inadequate. Target users: TfNSW Rail Operations Centre trainers, incident managers, emergency planners. The medium: the CBD model highlights the affected corridor in red and pulses the spread of delay minute by minute; interchange stations glow to show transferring passengers stranded. Interaction: an operator can "inject" a fault at a chosen junction via a control tablet and watch the modelled ripple. Data: GTFS-R Trip Updates and Service Alerts (live and archived), static GTFS for track topology, Opal patronage to weight passenger impact. Analysis: a delay-propagation model over the rail graph, using historical Trip Update deltas to estimate knock-on timings. Feasibility: moderately ambitious; the propagation model needs calibration, but archived GTFS-R makes validation possible.

3. Last Ferry, Last Train: Night Network Coverage A dusk-to-dawn time-lapse on the CBD model showing how the network thins after the evening peak, when the last services leave each corridor, and where NightRide buses do and do not fill the gap. Tackles night-time and off-peak service gaps affecting shift workers, who submissions to the Western Sydney inquiry noted cannot get trains after midnight and before 4am. Target users: TfNSW night-network planners, hospitality and health-sector employers, accessibility and equity advocates. The medium: the model dims to represent night; remaining services are projected as sparse moving lights, and corridors with no service go dark. Interaction: a physical dial scrubs time from 6pm to 5am. Data: GTFS-R Vehicle Positions and Trip Updates (to show real last-service timing versus schedule), static GTFS for NightRide routes, Opal Patronage for demand context. Analysis: computing service headways and coverage gaps by time-of-night and comparing scheduled versus actual last departures. Feasibility: achievable; largely a visualisation and scheduling-analysis task.

4. Interchange Choreography: Missed-Connection Detector Focuses on the CBD's dense interchanges (Central, Town Hall, Wynyard, Circular Quay, Barangaroo) and visualises, in real time, when a late feeder service causes passengers to miss a connecting service. Tackles interchange and transfer reliability and coordination between modes. Target users: TfNSW timetable planners, operators, the Rail Operations Centre. The medium: interchange nodes on the model light up amber when an arriving service is late enough to break a defined connection, and a projected counter tallies broken connections. Interaction: visitors select an interchange to see its connection success rate. Data: GTFS-R Trip Updates and Vehicle Positions across modes, static GTFS transfers and Pathways (walking times between platforms). Analysis: matching arrival and departure pairs at interchanges against pathway walk times to classify each connection as made or missed. Feasibility: moderate; requires careful modelling of realistic transfer windows using the Pathways extension.

5. George Street Squeeze: Light Rail versus Traffic A close focus on the George Street corridor showing CBD and South East Light Rail vehicles against live road conditions, highlighting where trams bunch or slow. Tackles light rail crowding and passenger-comfort complaints and the bus-versus-traffic interaction. Target users: City of Sydney, TfNSW active-transport and CBD coordination teams. The medium: the George Street spine is projected with tram positions and a heat colour for headway regularity; adjacent streets show live traffic hazards. Data: GTFS-R Vehicle Positions and Trip Updates for light rail, Live Traffic Hazards API, static GTFS. Analysis: headway-variance and bunching detection along the corridor. Feasibility: achievable and tightly scoped to one corridor.

Setup B: The Kinect AR sandbox (reshapeable terrain)

6. Terrain of Access: Reshape the City, See the Reach Users mould sand into a landscape while the projection overlays a real transport network and shows, as contour-coloured "reachability" bands, how far you can travel from a chosen point within 30, 45 and 60 minutes using live services. Tackles transport disadvantage and first mile / last mile problems by making accessibility tangible. Target users: community consultation participants, students, planning students, equity advocates. The medium: the Kinect reads the sand surface; a user places a marker (a hand or object) to set an origin, and the projection paints live isochrones that shift as real services run. Reshaping the sand into "hills" can represent barriers such as rivers or motorways to discuss severance. Data: GTFS-R Trip Updates plus static GTFS to build a time-dependent network, Trip Planner API for validation, Opal Patronage for context. Analysis: real-time isochrone (travel-time catchment) computation using current service timings rather than the timetable, echoing published rt2gtfs and realizable-accessibility research. Feasibility: ambitious; real-time isochrone computation is heavy but well precedented, and the sandbox mapping is the mature AR-sandbox pattern.

7. Flood and Fail: Resilience Sandbox Users sculpt terrain and "pour" a flood or place an incident, and the system shows which rail and bus corridors would be cut and which communities lose access, referencing real events such as the Blue Mountains landslide and the flood-prone rail zones named in the Rail Review. Tackles network resilience and the drainage focus of the Independent Rail Review. Target users: emergency planners, SES liaison, TfNSW resilience teams, councils. The medium: the Kinect senses where the user raises or lowers

terrain and where they gesture to place water; corridors crossing flooded cells turn red and dependent stops flash. Data: GTFS-R Service Alerts and Trip Updates (live disruption), static GTFS, Live Traffic Hazards flood and incident feeds. Analysis: overlaying a user-defined hazard footprint on the network graph and recomputing reachable stops and affected patronage. Feasibility: moderately ambitious; the hazard-to-network intersection is straightforward, and live flood alerts add realism.

8. Grade of Access: Step-Free Sandbox A sandbox where raising terrain represents stairs and level ground represents step-free access, letting users feel the difference between an accessible and inaccessible journey while the projection routes a wheelchair icon around barriers using real service availability. Tackles accessibility and step-free access gaps (for example the single-lift risk at stops like Lilyfield, where one lift out of service makes the stop inaccessible). Target users: accessibility advocates, TfNSW Transport Access Program teams, DSAPT compliance staff. The medium: the Kinect reads raised "step" zones; the projected accessible route avoids them and reroutes live if a lift is out (from Service Alerts). Data: GTFS-R Service Alerts and Trip Updates, GTFS Pathways, DSAPT compliance dataset, facilities and accessibility data, vehicle descriptor wheelchair flags. Analysis: accessible-only shortest path that responds to live lift-outage alerts and wheelchair-accessible vehicle availability. Feasibility: moderate; the accessibility routing is the substance and the sandbox is a strong engagement device.

9. Demand Dunes: Patronage and Crowding Landscape The projection renders a live "crowding landscape" over the sand where corridors rise in colour intensity with load, and users can flatten peaks by hand to explore what added capacity would relieve them. Tackles overcrowding on corridors like the T1 Western at 145% morning load. Target users: TfNSW network planners, students. The medium: the projection shows crowding heat over the network on the sand; when a user lowers a "peak" the system shows the extra services needed to hit the 135% benchmark. Data: Sydney Trains v2 predictive occupancy feed, GTFS-R Trip Updates, Opal Patronage. Analysis: mapping live occupancy to load factors and computing the service frequency required to bring loads under benchmark. Feasibility: moderately ambitious; depends on the v2 occupancy feed, which currently covers Sydney Trains.

Setup C: The LEGO participatory planning grid (stereo-photogrammetry / homography input)

10. Build the Bus Route: Participatory Network Designer Community members place coloured LEGO bricks on a gridded map of a growth area (for example the Aerotropolis or a Western Sydney suburb) to propose bus routes and stops; the system reads the layout and instantly scores it against live demand and existing services. Tackles Western Sydney transport deserts and the inquiry call to plan transport before housing. Target users: community consultation participants, local councils, WSROC, TfNSW network planners. The medium: brick colour encodes route or stop type and height encodes frequency; stereo-photogrammetry or homography captures the layout; the projection overlays the resulting coverage and gaps. Interaction loop: place bricks, see coverage and estimated catchment update, adjust. Data: static GTFS and GTFS-R (existing service baseline), Opal Patronage and Tap On/Tap Off origin-destination by travel zone, facilities data. Analysis: catchment and coverage scoring, comparing the proposed network against real demand and current gaps. Feasibility: achievable for the input capture; the scoring model is the substantive work. A strong consultation tool.

11. Where Should the Station Go: TOD Trade-off Table Participants place a LEGO "station" brick and surrounding "density" bricks along a corridor such as Metro West (Parramatta to Sydney Olympic Park) and the system shows the trade-off between homes served, jobs reached and service load, echoing the inquiry recommendation to consider additional Metro West stations. Tackles transport-oriented development and matching transport supply to housing growth. Target users: councils, planners, TOD program teams, community. The medium: the grid represents the corridor; station bricks add a stop to the modelled line and density bricks add dwellings; the projection shows changes in catchment population and live travel times to the CBD. Data: GTFS-R and static GTFS for the corridor, Opal Patronage, facilities and accessibility data, cycleway data for active-transport access. Analysis: accessibility and catchment computation per candidate station configuration. Feasibility: moderately ambitious; requires a corridor model but delivers clear planning insight.

12. Green the Fleet: Zero Emission Depot and Route Planner Participants use LEGO to place depots and assign routes, and the system shows which routes a battery-electric fleet could serve given range and charging, and the emissions saved, tied to the 2035 Greater Sydney ZEB target. Tackles the zero emission bus transition and depot electrification planning. Target users: bus operators, TfNSW ZEB program, councils, students. The medium: depot bricks and route bricks on a regional grid; the projection colours routes by electrification feasibility and shows fleet and charging implications. Data: static GTFS and GTFS-R (route lengths, running times, live cycle times), depot locations, patronage. Analysis: matching route duty cycles against battery range and charging windows, and computing emissions reductions. Feasibility: moderate; the range-and-duty-cycle modelling is well understood and the LEGO input makes scenarios tangible.

Recommendations

1. **Start with Project 1 (Pulse CBD) as the flagship.** It is the most achievable, exercises the full GTFS-R ingestion stack for every mode, and calibrates the CBD model projection, which is reusable by Projects 2, 3, 4 and 5. Build it first and treat it as shared infrastructure. Benchmark to move on: a stable multi-mode feed ingester running at the 15-second cadence with map-matched positions.
2. **Pair one ambitious analytical project with each setup** so the portfolio demonstrates range: Cascade (Project 2) on the model, Terrain of Access (Project 6) on the sandbox, and Build the Bus Route (Project 10) on the LEGO grid. These three carry the strongest problem narratives: reliability cascades, accessibility equity, and Western Sydney deserts.
3. **Use archived GTFS-R for anything involving cascades, crowding or history.** Live feeds alone cannot show the Homebush event or validate a propagation model; capture and store the feeds now, following the SCALUT and Nature Scientific Data pipeline pattern, so replay and validation are possible. Benchmark: at least a few weeks of stored Trip Updates and Vehicle Positions before attempting Project 2 or 9.
4. **Secure an API key and confirm the v2 occupancy feed early** if pursuing crowding (Project 9) or accessibility vehicle flags (Project 8), because these depend on the extended feeds and the proto file. If the v2 occupancy feed proves too limited, fall back to Opal Patronage-driven typical-day crowding rather than live occupancy.

5. **For consultation projects (10, 11, 12), co-design with an actual stakeholder** such as a Western Sydney council or WSROC, so the scoring reflects real planning criteria and the output can feed a genuine Have Your Say process. The threshold that should change the approach: if a partner cannot commit, build these as museum or education pieces with illustrative rather than decision-grade scoring.
6. **Sequence by effort.** Achievable now: Projects 1, 3, 5, 10. Moderate: 4, 7, 8, 11, 12. Ambitious: 2, 6, 9. Deliver two achievable projects to prove the pipeline before committing to an ambitious one.

Caveats

- On-time running figures use TfNSW definitions that exclude cancelled, incomplete and untracked trips as well as contract-agreed exclusions, so headline punctuality can understate the passenger experience of cancellations. Bus cancellation and tracking data has its own gaps, with untracked trips recorded as a distinct category.
- Opal patronage methodology changed from 1 July 2024, and line, agency and mode aggregations are no longer valid to combine; tap-level origin-destination by travel zone exists only for four weeks of 2020, so it is illustrative rather than current.
- GTFS-R feeds have documented quirks (ferry position trails, changing bus trip IDs, ad hoc light rail updates, regional feed defects) that any project must handle defensively.
- Live occupancy and carriage-level prediction are strongest for Sydney Trains via the v2 feed; equivalent live crowding for buses and other modes is more limited.
- Several cited items are projections or plans (Metro West 2032, the Aerotropolis employment mix, the ZEB 2035 target) rather than delivered outcomes, and timelines have slipped before (the Bankstown conversion moved from 2024 to 2026).
- A small number of patronage rankings and station figures rest on secondary reprocessing of TfNSW data rather than primary TfNSW publications, and are directionally reliable rather than exact.